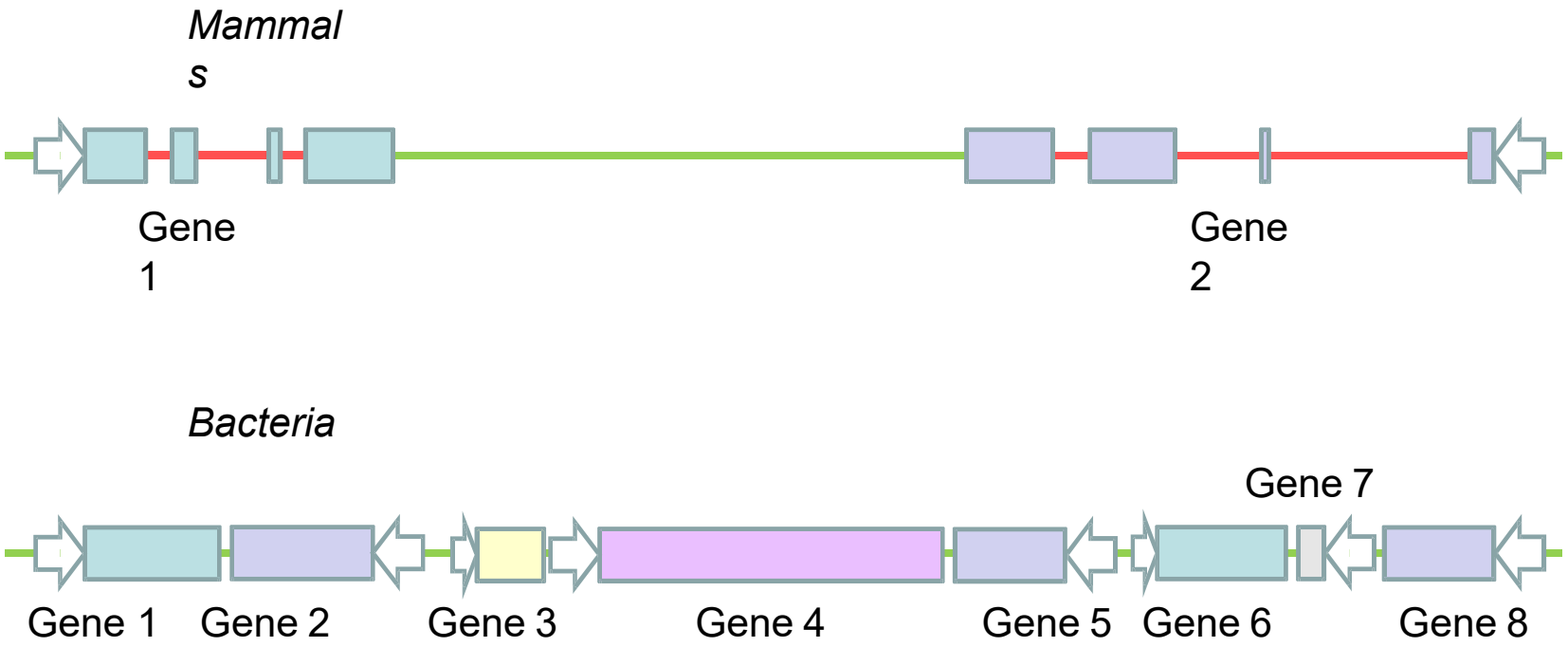
Chapter IV: Genetic mutation
Genetic variation, genetic markers.

Genetic Mutations

Genomes and kingdom

Organism	Genome size (one copy)	Nb of genes
Bacteria	4000 kb	4000
Eukaryotes: Yeast	12000 kb	6000
Drosophila	137 Mb	14000
Rice	466 Mb	40000
Mammal	3 Gb	30000
Homo sapiens	3 Gb	30000
Wheat	17 Gb	

Genomes et kingdoms



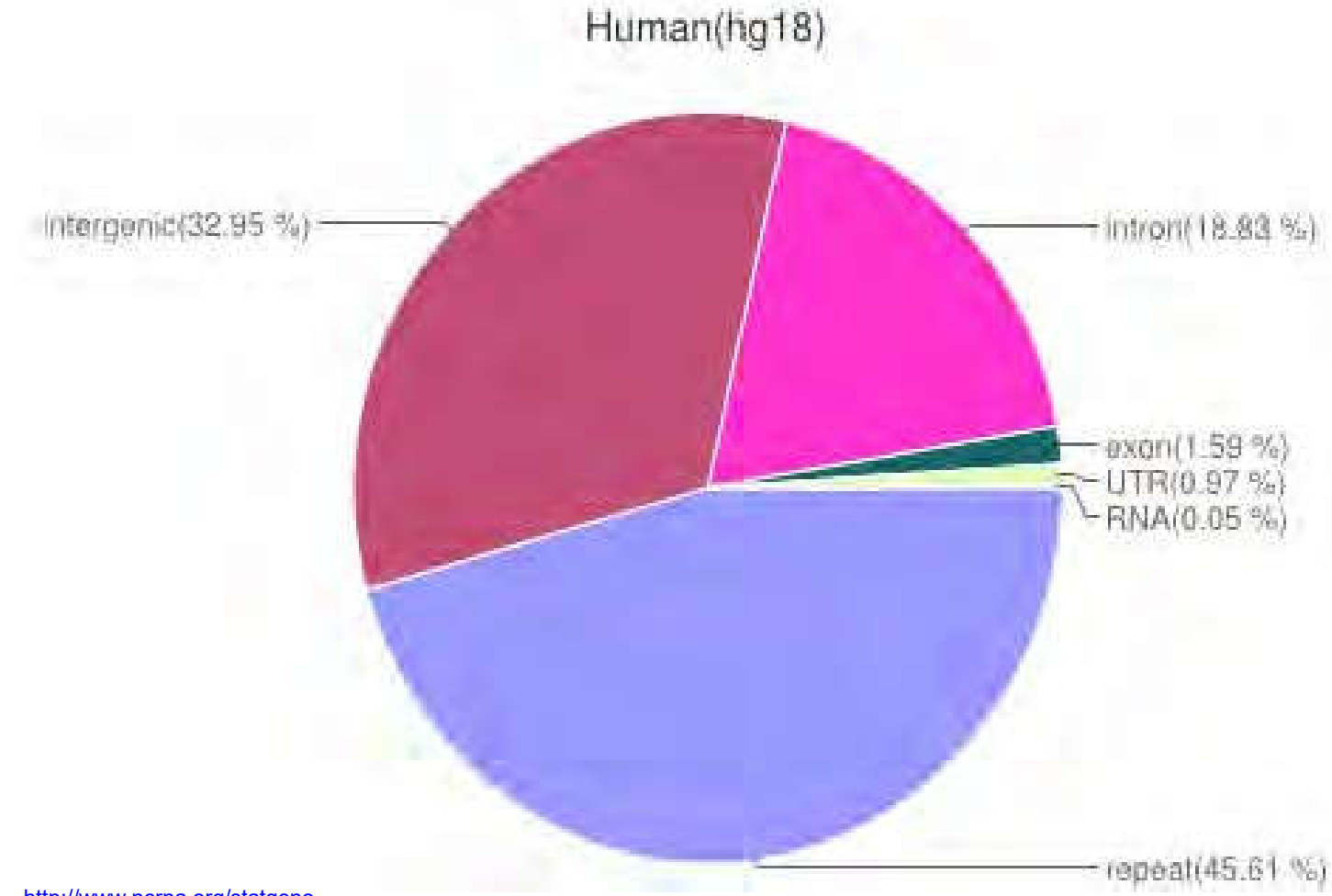
Genomes and kingdom

In higher eukaryotes (plants, animals),

genes tend to be larger
mostly because they contain introns

and thay are separated by intergenic sequences.

Only a small part of the human genome is translated

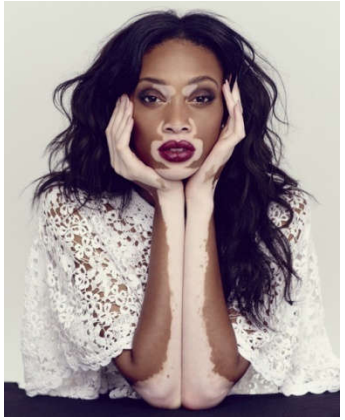


http://www.ncrna.org/statgenome

What Are Mutations?

- Changes in the nucleotide sequence of DNA
- May occur in somatic cells (aren't passed to offspring)
- May occur in gametes (eggs & sperm) and be passed to offspring

What Are Mutations?



- Mutations happen regularly

  Virus: 10^{-6} ~ 10^{-8} mutations /base /generation (DNA)

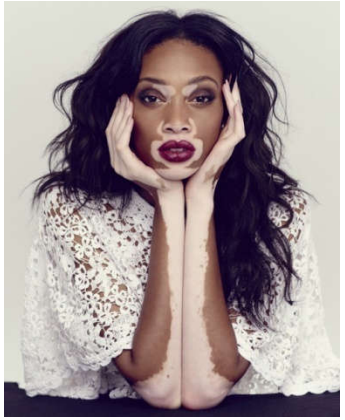
  10^{-3} to 10^{-5} per base per generation (RNA)

  Human: 10^{-8} mutations /base /generation

- Almost all mutations are neutral

- Chemicals & UV radiation cause mutations

What Are Mutations?



- Many mutations are repaired by enzymes:
  Human mutation rate is close to 1.0×10^{-8}. Many errors are repaired by repair enzymes and this process is about 99% efficient. Thus, the overall error rate is close to 1.0×10^{-10} per bp.
- Some type of skin cancers and leukemia result from somatic mutations
- Some mutations may improve an organism's survival (beneficial)

mutagens

Class	Mutagen
Physical mutagens 1.Ionizing radiations -particulate radiations -non-particulate radiations 2.Non-ionizing radiations	 -Alpha ,beta ,fast neutrons, thermal neutrons -X-rays, gamma rays -UV rays
Chemical mutagens 1.Alkylating agents -base analogues -acridine dyes 2.Deamination agents -other chemical mutagens	-mustard gas, nitrogen mustard, EMS,MMS,EES. -5-bromouracil,2-amino purine -acriflavin,proflavin,acridine orange. -nitrous acid (HN02) -hydroxylamine, sodium azide. EMS-ethylmethane sulphonate, MMS-methylmethane sulphonate, EES-ethylene sulphonate.

List of genetic disorders

Disorder	Mutation	Chromosome
22q11.2 deletion syndrome	D	22q
Color blindness	P	X
Cri du chat	D	5
Cystic fibrosis	P	7q
Klinefelter syndrome	C	X
Neurofibromatosis		17q/22q/?
Phenylketonuria	P	12q
Polycystic kidney disease	P	16 (PKD1) or 4 (PKD2)
Prader–Willi syndrome	DC	15
Sickle-cell disease	P	11p
Spinal muscular atrophy	DP	5q
Tay–Sachs disease	P	15
Turner syndrome	C	X

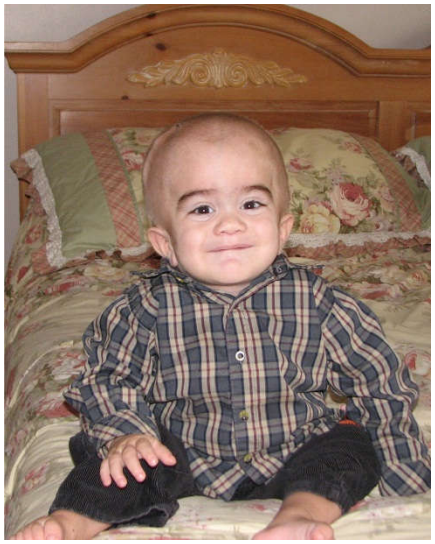
- P – Point mutation, or any insertion/deletion entirely inside one gene
- D – Deletion of a gene or genes
- C – Whole chromosome extra, missing, or both (see Chromosome abnormality)

Gene Mutation Associated with Intellectual Dysfunction

- Ogden Syndrome, also known as n-terminal acetyltransferase deficiency (NATD), comprising a distinct combination of distinctive craniofacial features producing an aged appearance, growth failure, hypotonia, global developmental delays, cryptorchidism, and acquired cardiac arrhythmias.

- Lyon and colleagues discovered Ogden syndrome, finding it in two generations of a family with five affected boys. All of the affected individuals died by the age of three. Ogden syndrome is an X-linked condition - only affecting males - and is caused by an error in the NAA10 gene.

- The researchers have gone on to get referrals from clinics around the globe, identifying 37 people in 32 families that harbor mutations in the NAA15 gene. The NAA15 gene is not on the X chromosome, so the group includes people of both sexes.



Odgen syndrome

Gene Mutation Associated with Intellectual Dysfunction



Types of Mutations

- Chromosome structure mutation:
- Gene mutation:

Chromosome Mutations

- May Involve:
  - **Changing the structure**

loss or gain

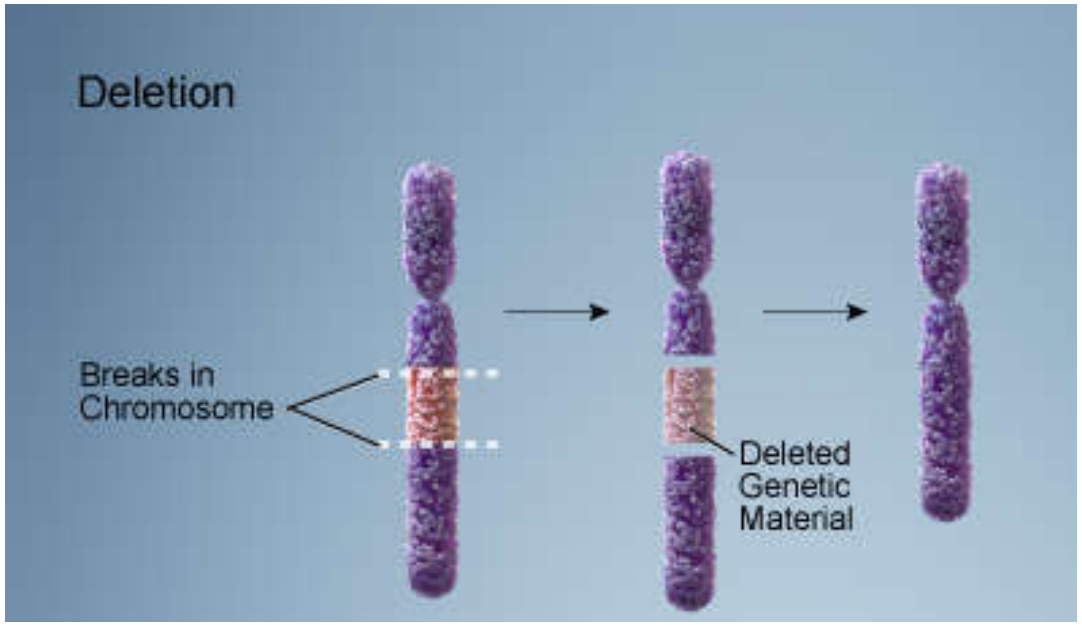


Chromosome Mutations

- Five types exist:
  - **Deletion**
  - **Inversion**
  - **Translocation**
  - **Nondisjunction**
  - **Duplication**

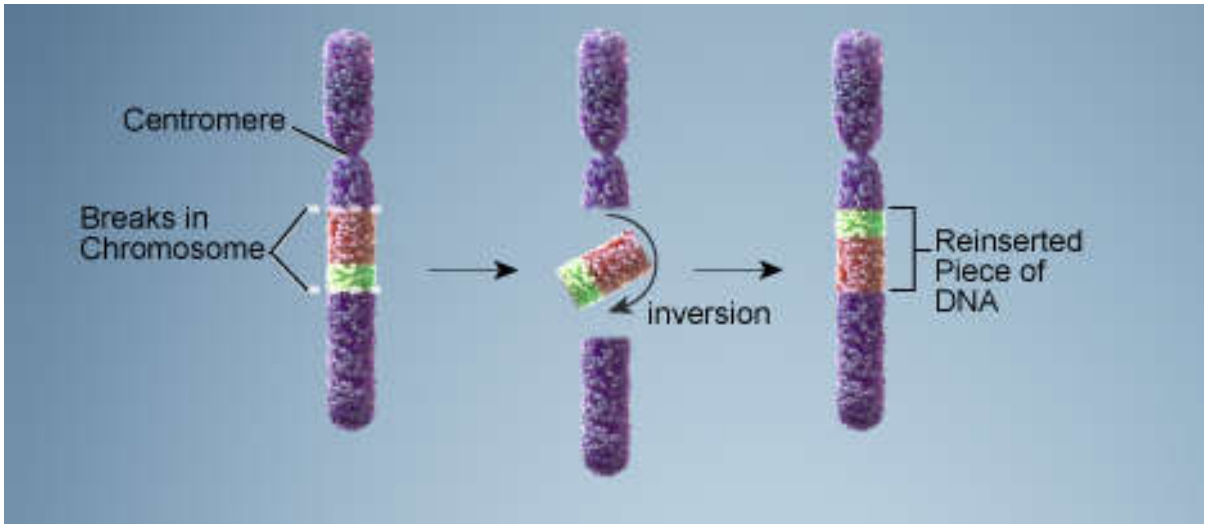
Deletion

- Due to **breakage**
- A **piece** of a chromosome is **lost**



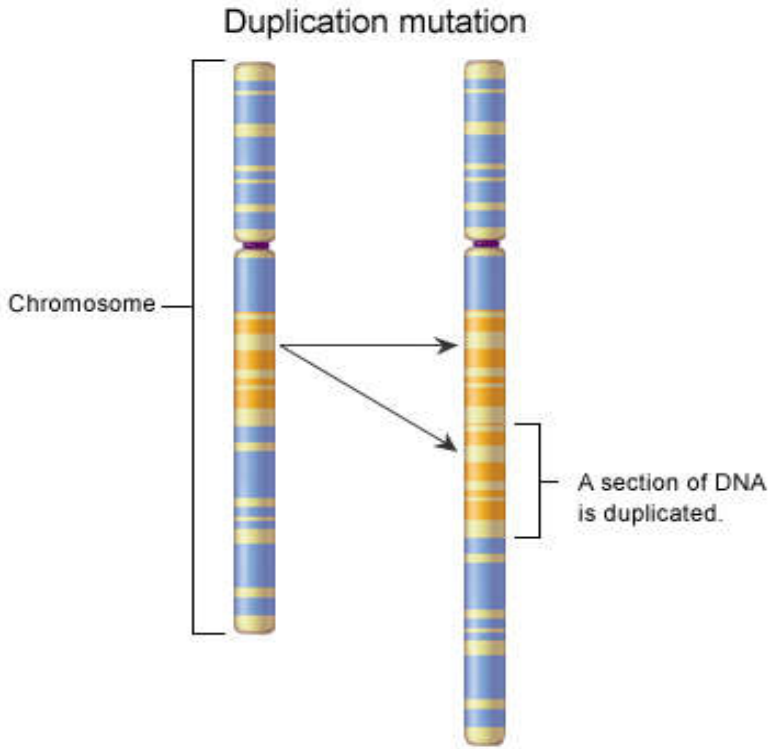
Inversion

- Chromosome segment **breaks off**
- Segment flips around **backwards**
- Segment **reattaches**



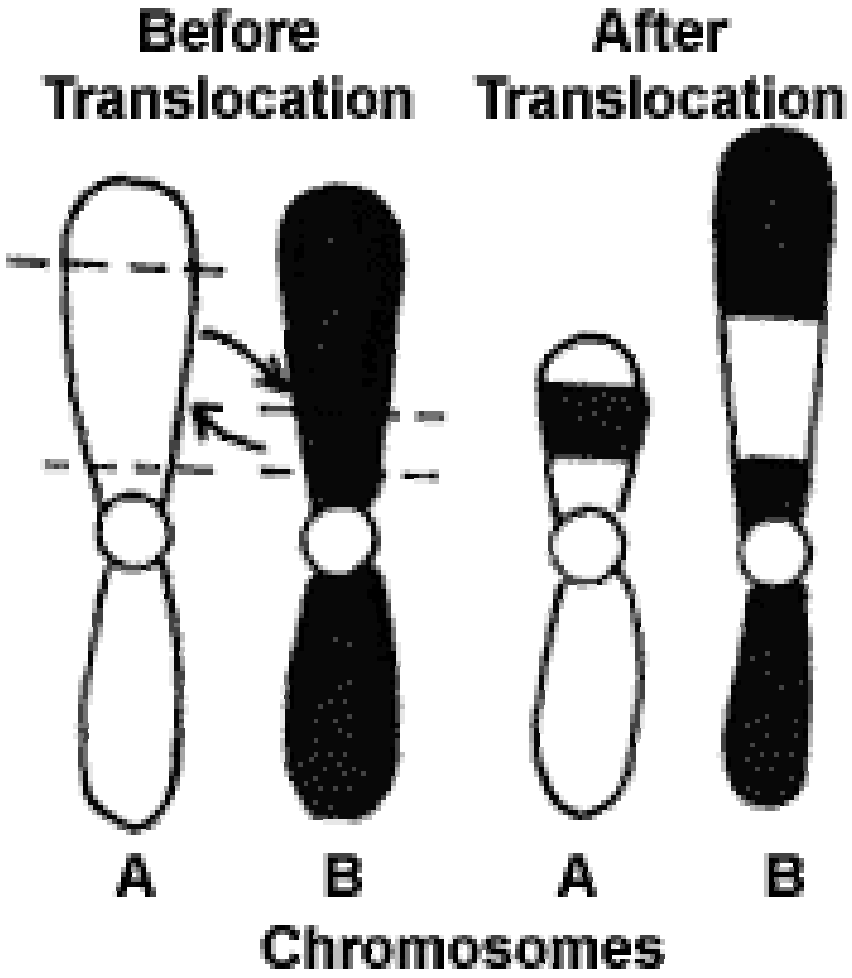
Duplication

- Occurs when a gene **sequence is repeated**

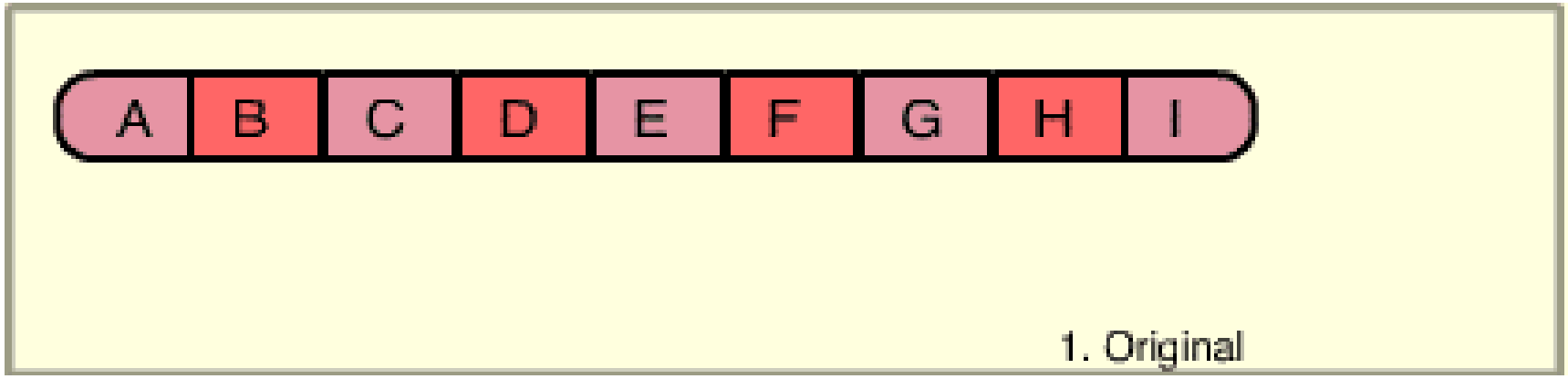


Translocation

- Involves **two chromosomes** that aren't homologous
- **Part** of one chromosome is **transferred to another** chromosomes

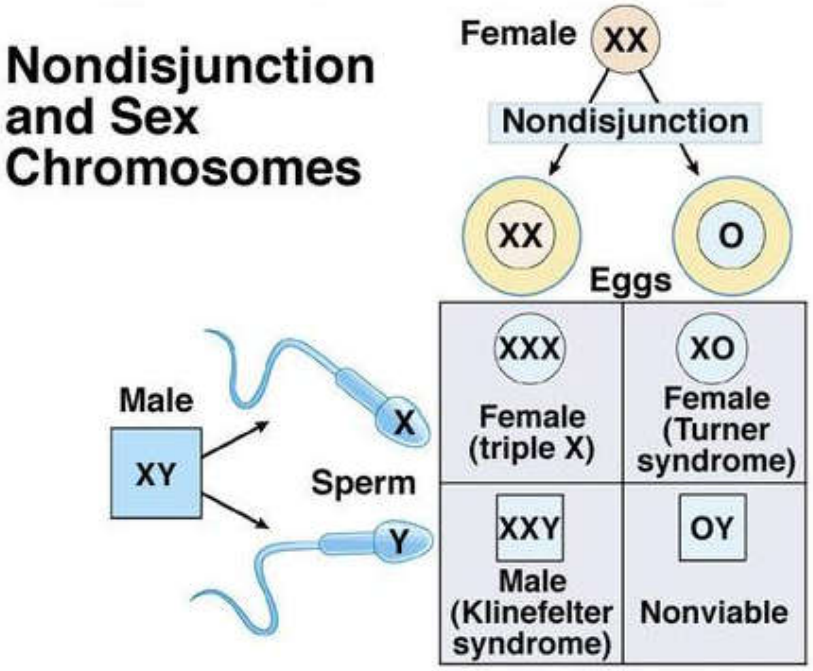


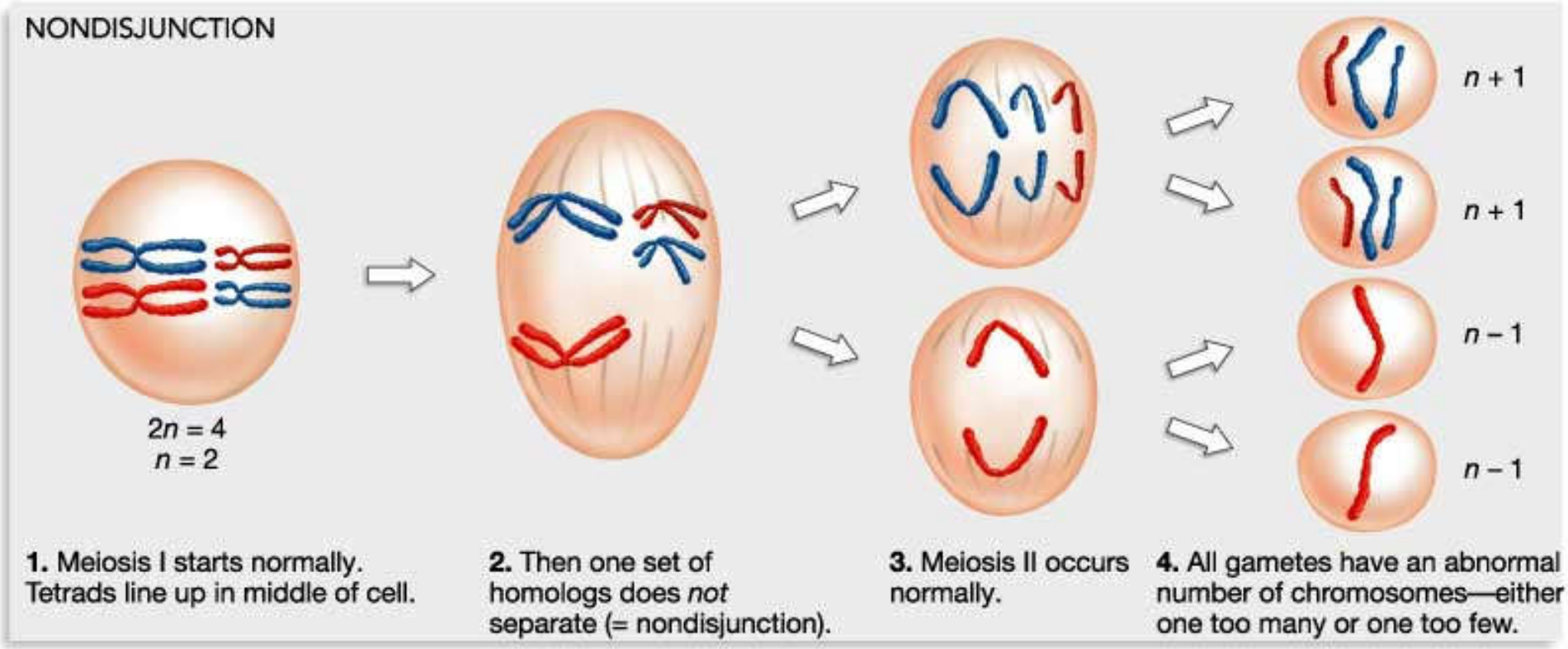
Chromosome Mutation Animation



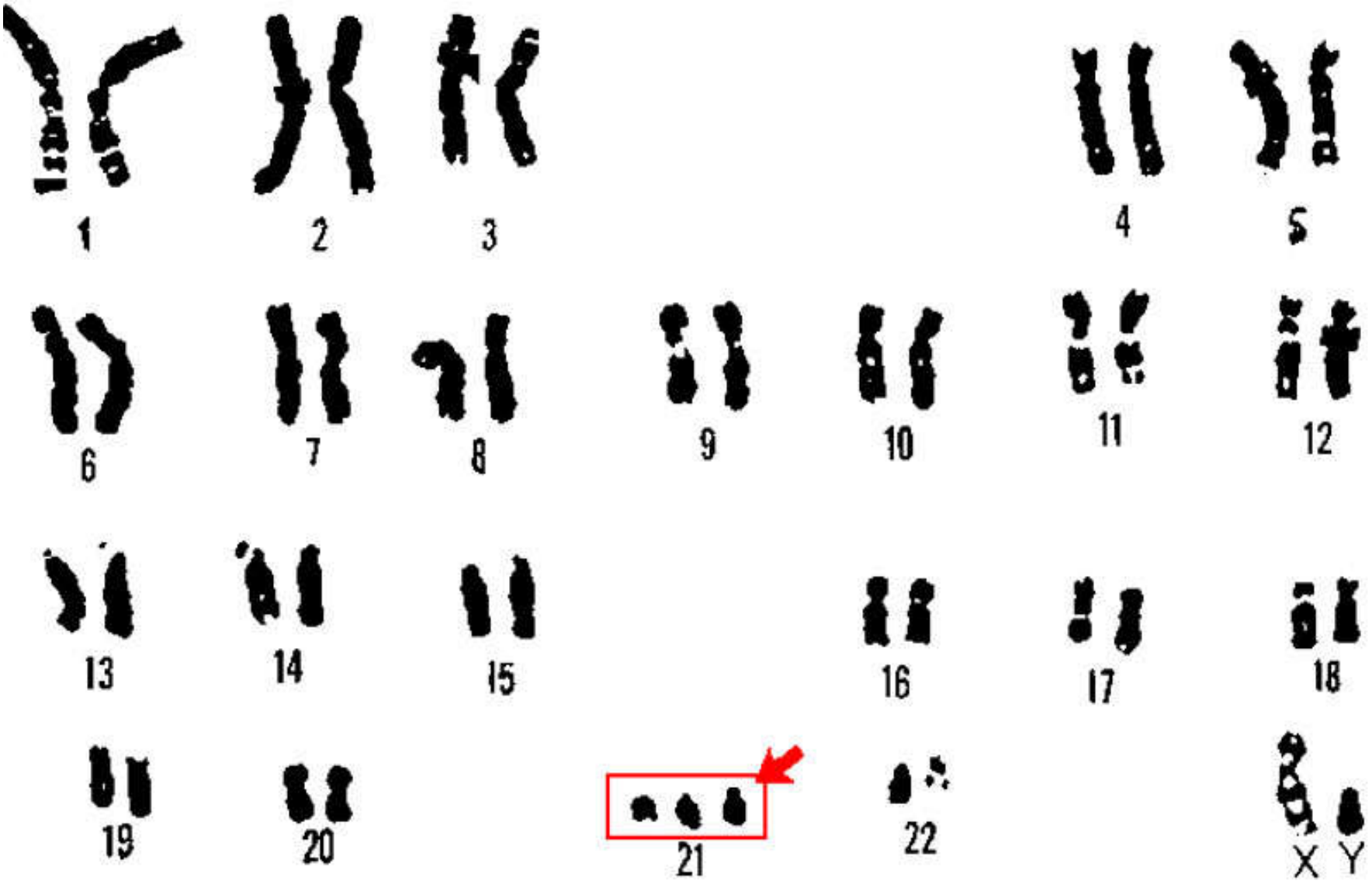
Nondisjunction

- **Failure** of chromosomes **to separate** during meiosis
- Causes gamete to have **too many** **or** **too few** **chromosomes**



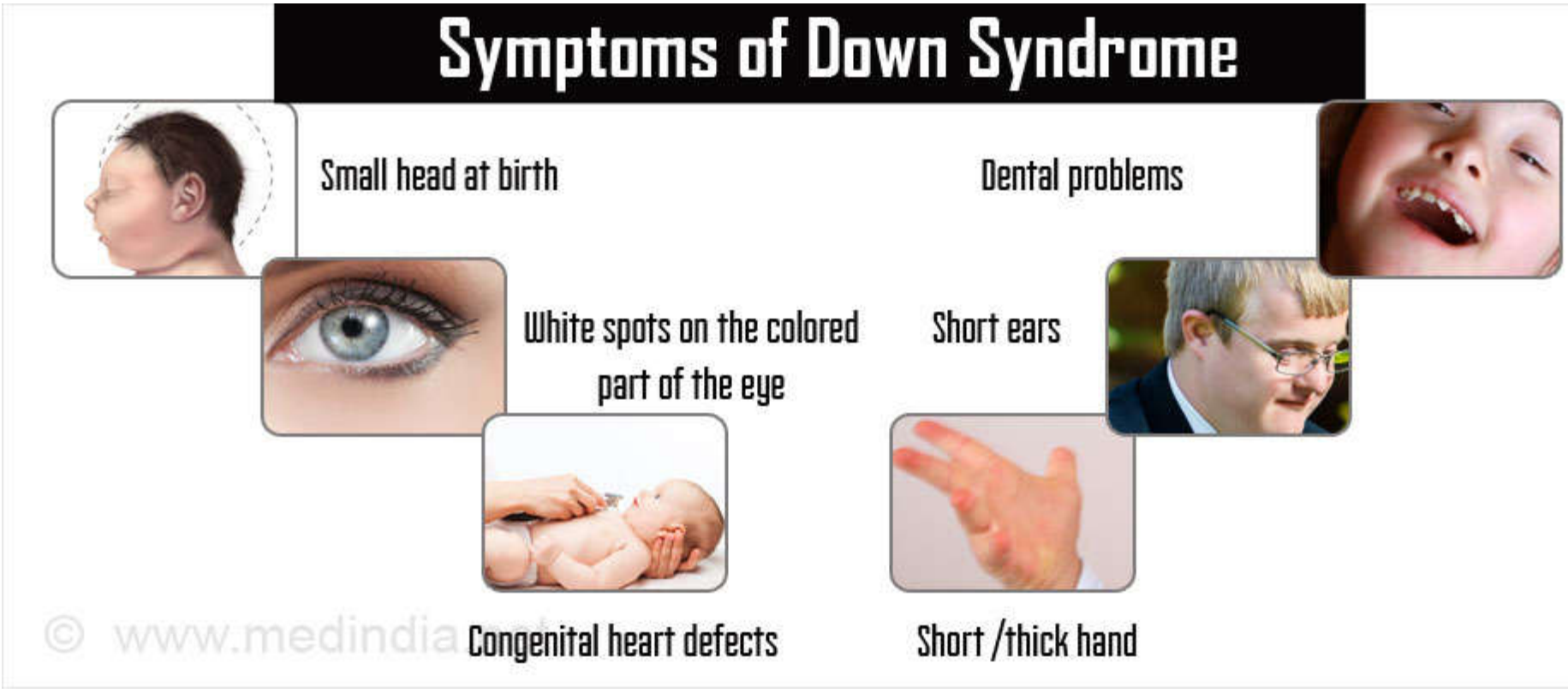

NONDISJUNCTION
2n = 4
n = 2
n + 1
n + 1
n – 1
n – 1
1. Meiosis I starts normally. Tetrads line up in middle of cell.
2. Then one set of homologs does not separate (= nondisjunction).
3. Meiosis II occurs normally.
4. All gametes have an abnormal number of chromosomes—either one too many or one too few.

Down syndrome

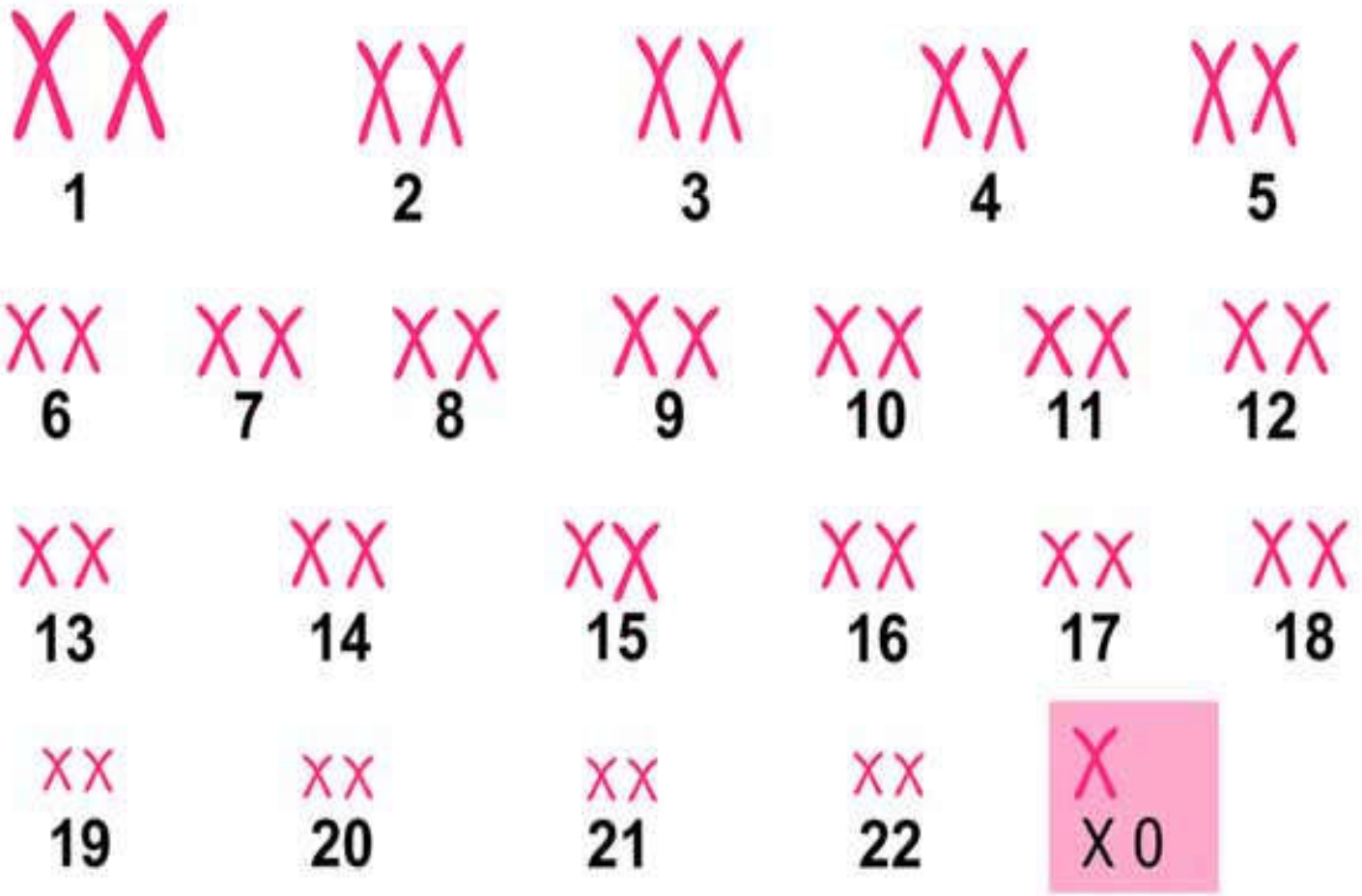


Down syndrome

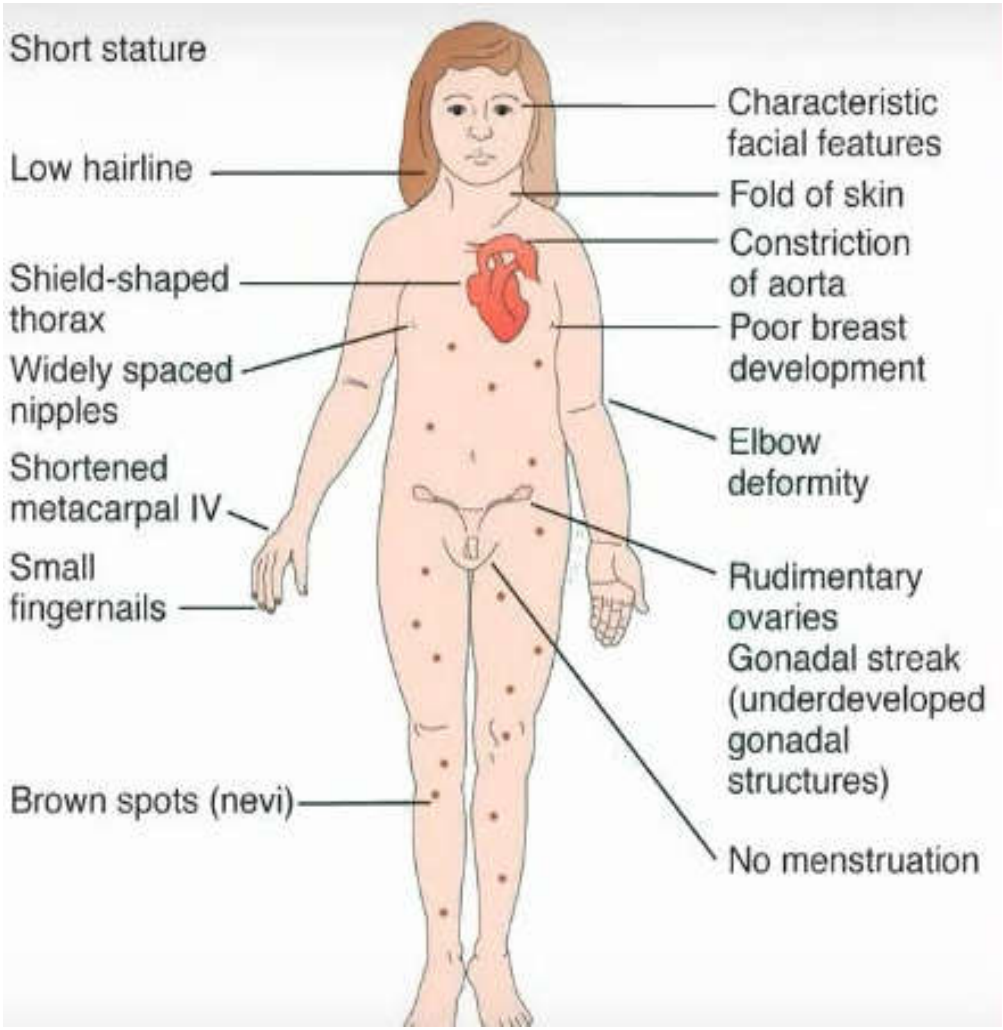




Turner syndrome karyotype



Turner syndrome

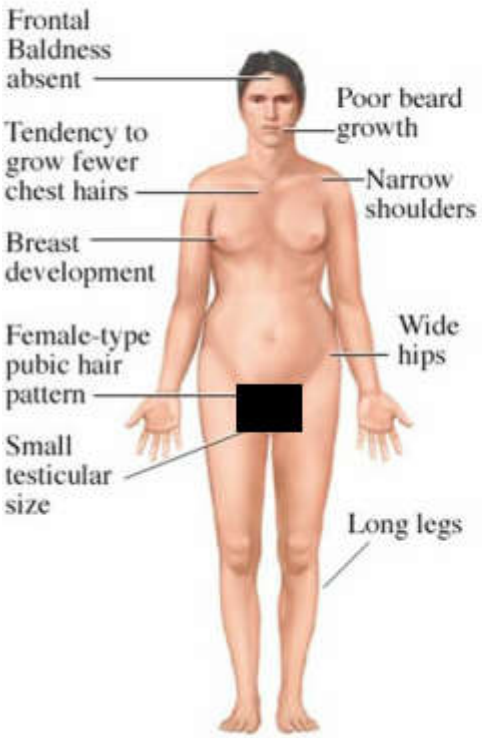




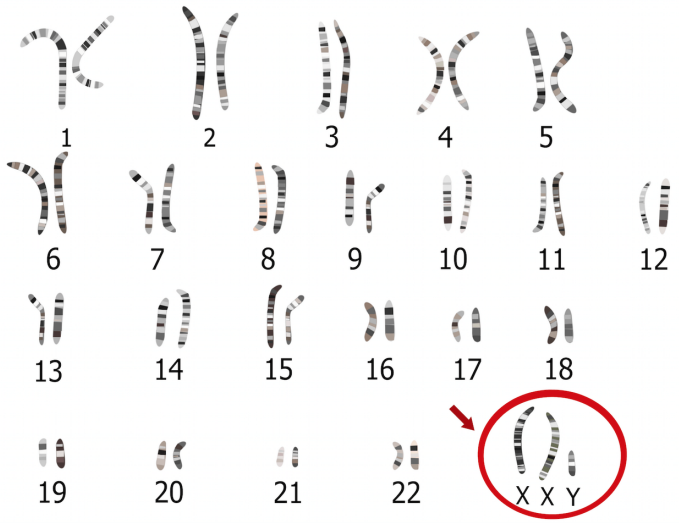
medgen.genetics.utah.edu

Klinefelter syndrome

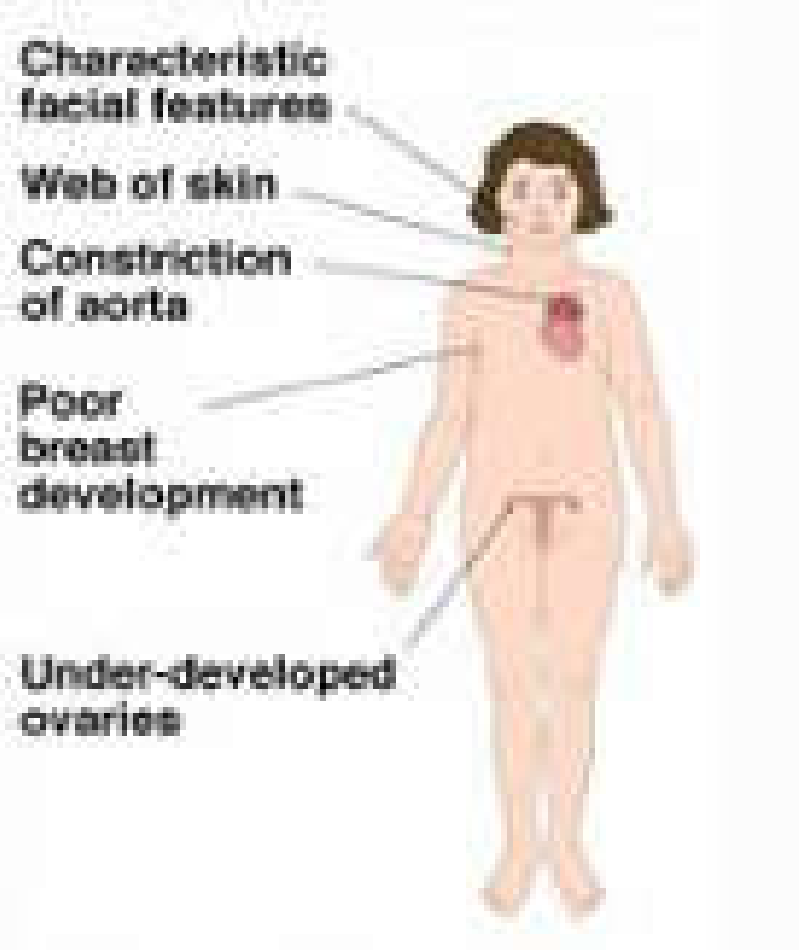
Klinefelter syndrome

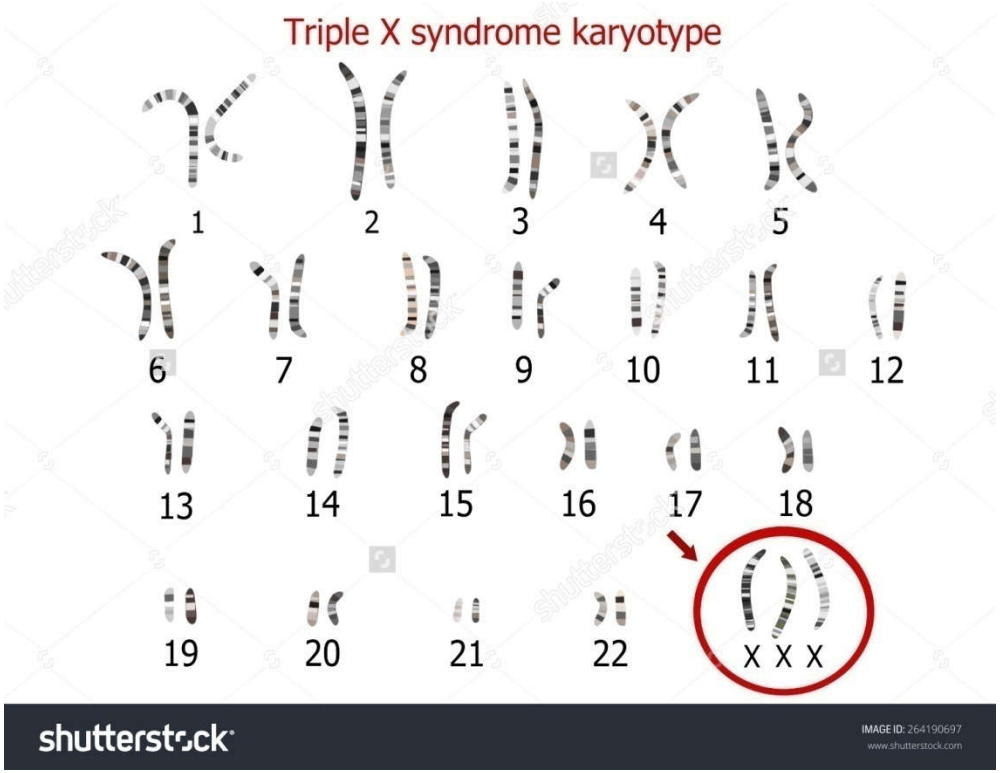


- Lower IQ than sibs
- Tall stature
- Poor muscle tone
- Reduced secondary sexual characteristics
- Gynaecomastia (male breasts)
- Small testes/infertility

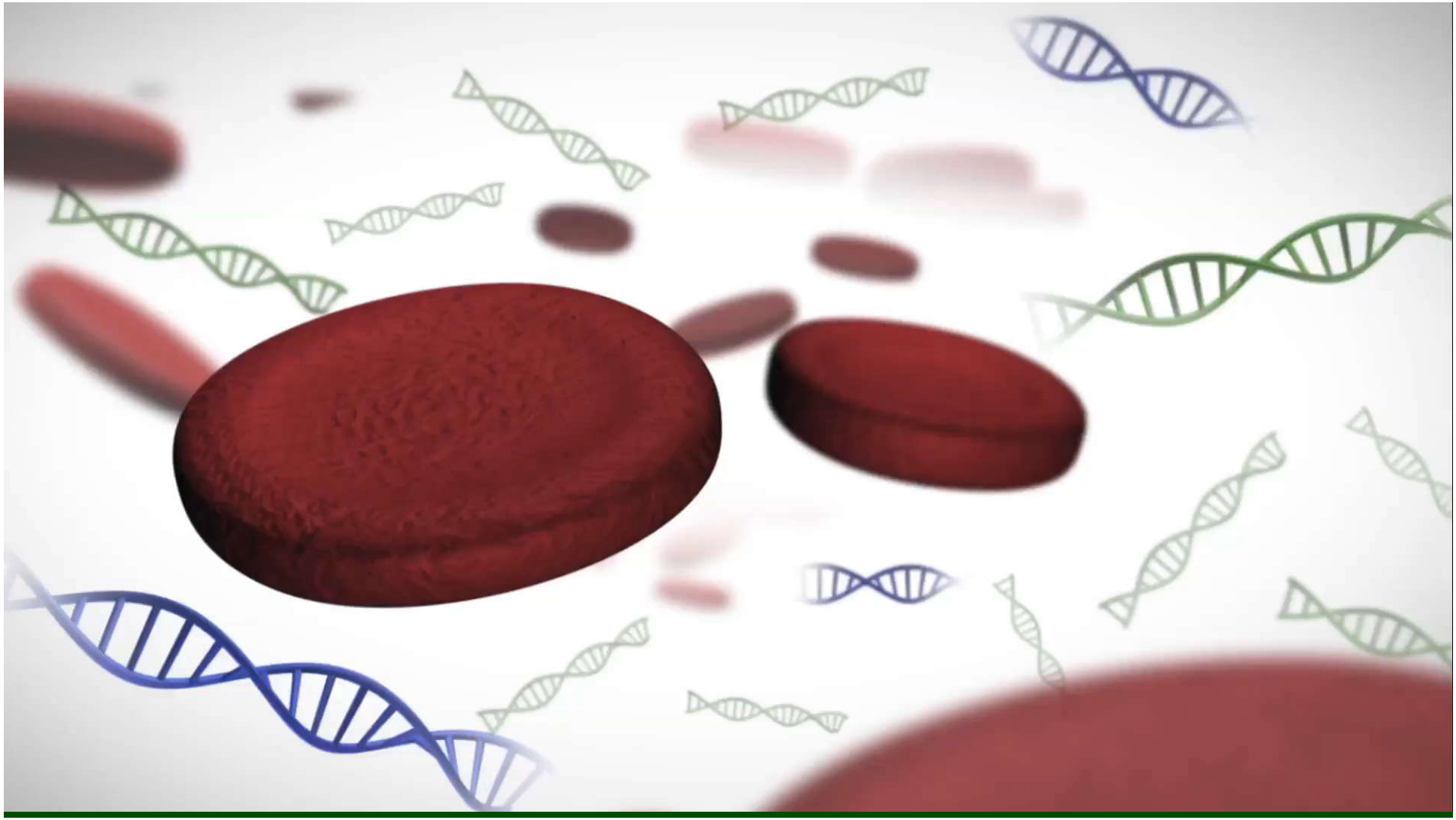


Triple X syndrome





NIPT (non invasive prenatal testing)



Gene Mutations

- Change in the **nucleotide sequence** of a gene
- May only involve a single nucleotide
- May be due to **copying errors**, **chemicals**, **viruses**, etc.

Types of Gene Mutations

- Include:
  - Point Mutation (substitution, deletion, insetion).
  - Insertion
  - Deletion
  - Frameshift

Point Mutation

- Change of a **single** nucleotide
- Includes the deletion, insertion, or substitution of **ONE** nucleotide in a gene

Point Mutation

- **Sickle Cell disease** is the result of one nucleotide substitution
- Occurs in the **hemoglobin gene**



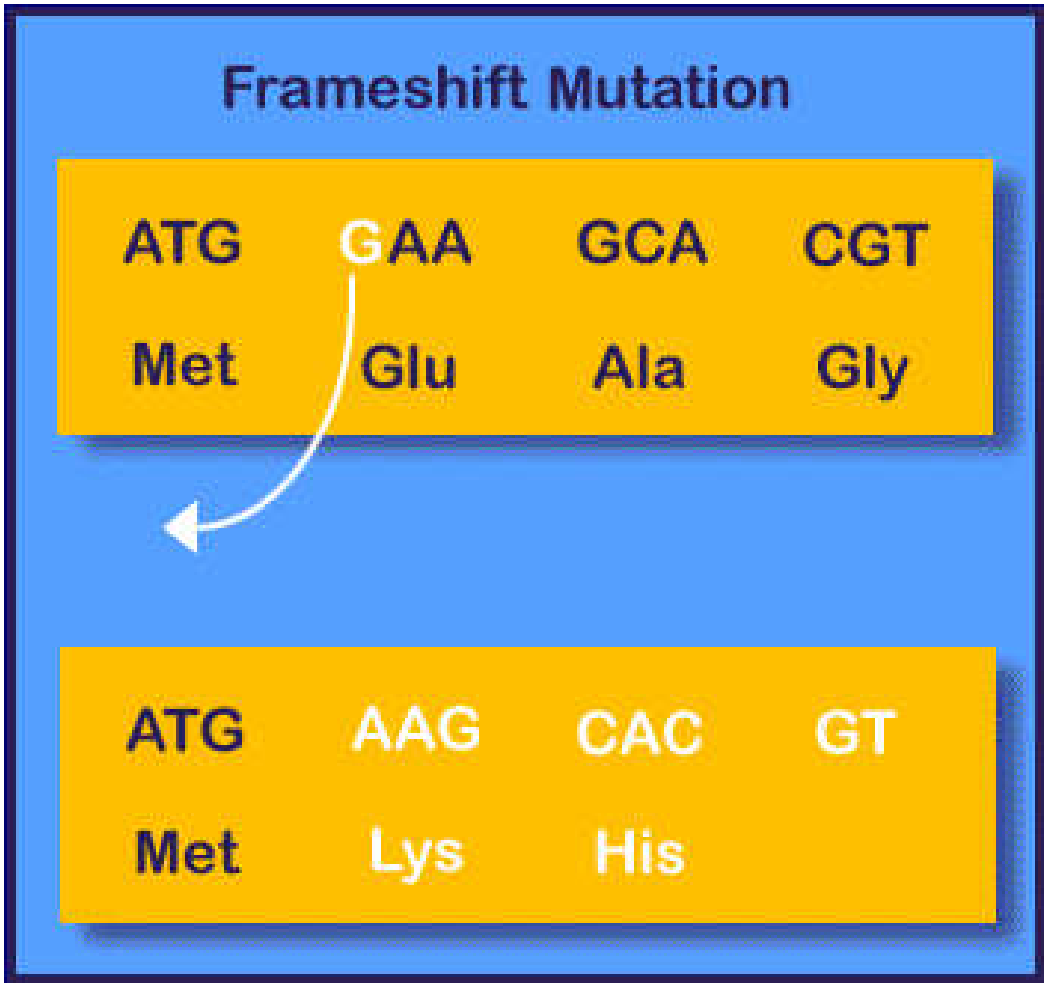
Frameshift Mutation

- **Inserting or deleting** one or more nucleotides
- Changes the "**reading frame**" like changing a sentence
- **Proteins** built **incorrectly**

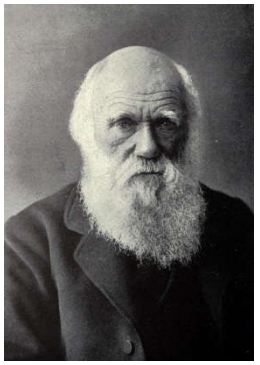
Frameshift Mutation

- Original:
  - The fat cat ate the wee rat.
- Frame Shift (“a” added):
  - **The fat caa tet hew eer at.**

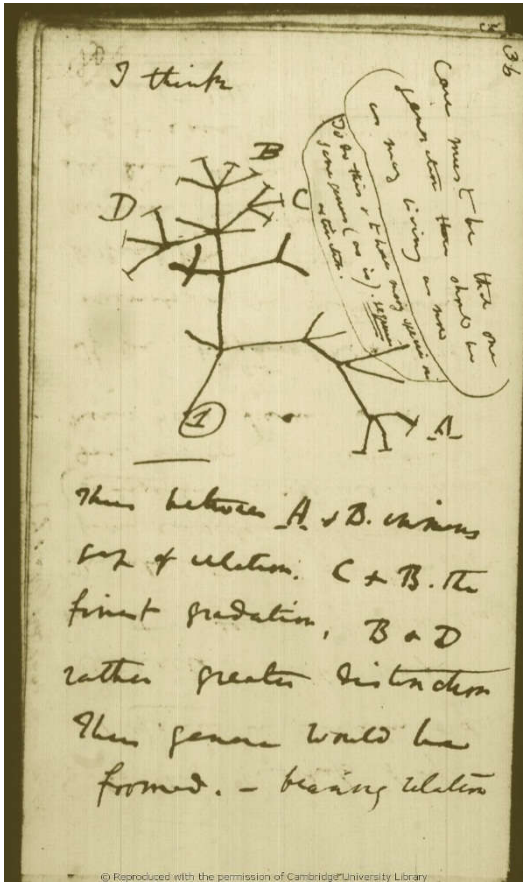
Amino Acid Sequence Changed



Genetic mutations are raw material of evolutions

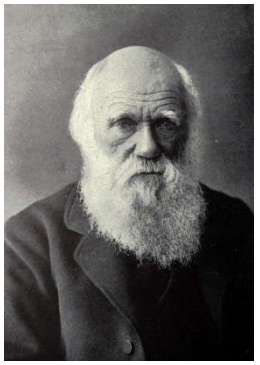


Charles Darwin
(1809-1882)

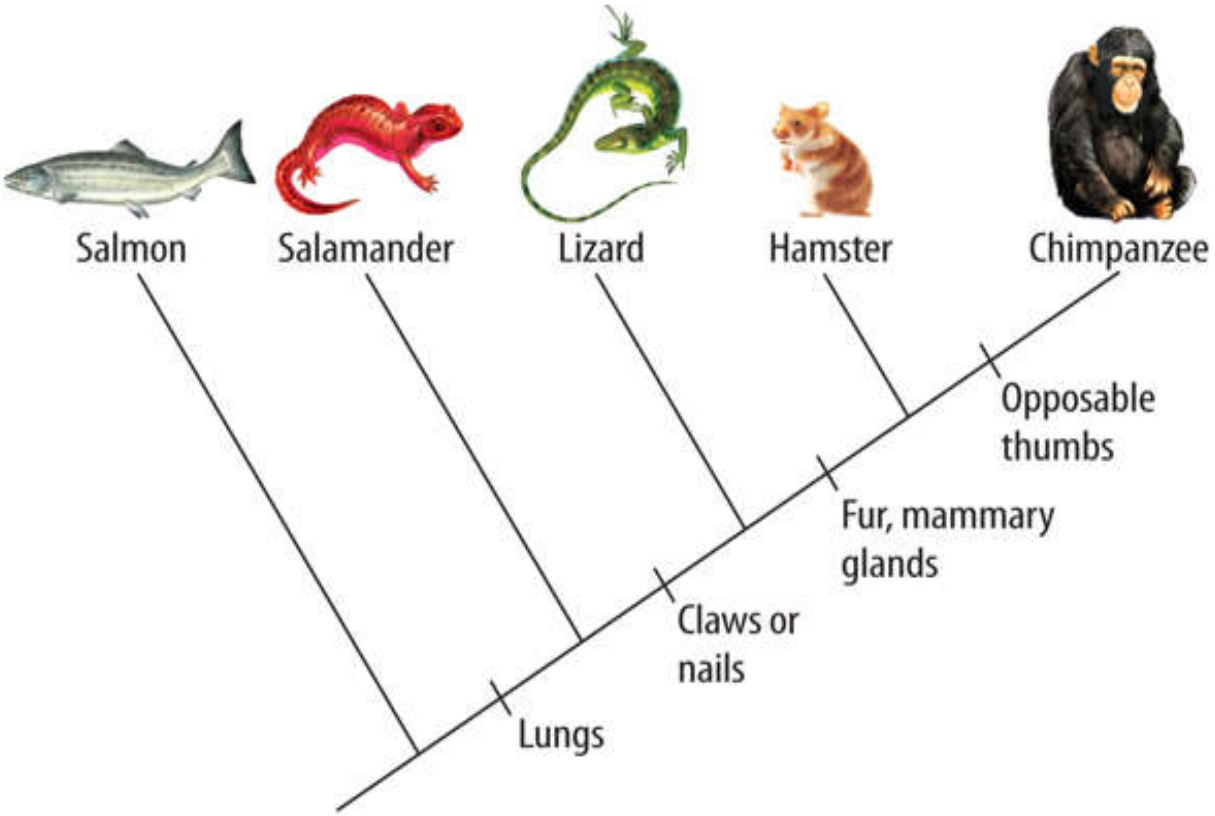


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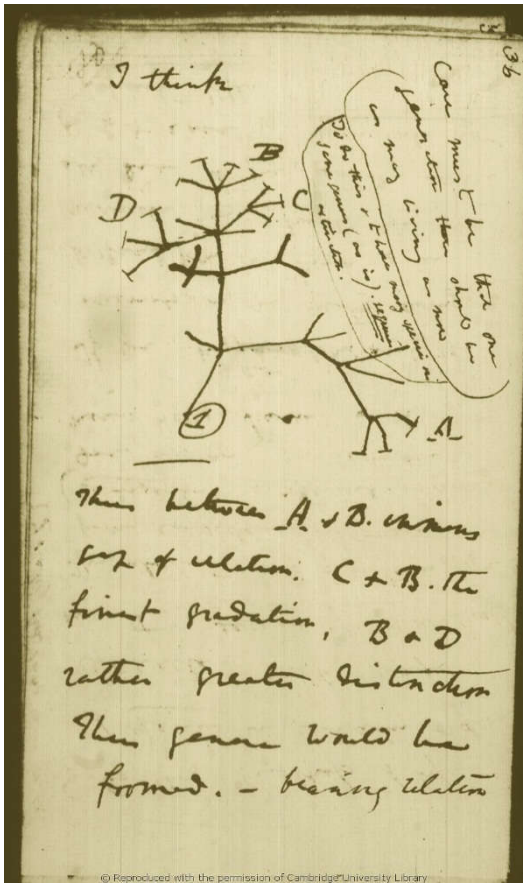
Genetic mutations are raw material of evolutions



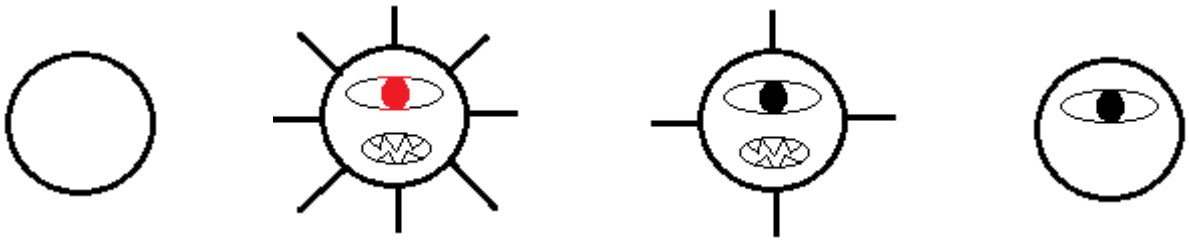
Charles Darwin
(1809-1882)



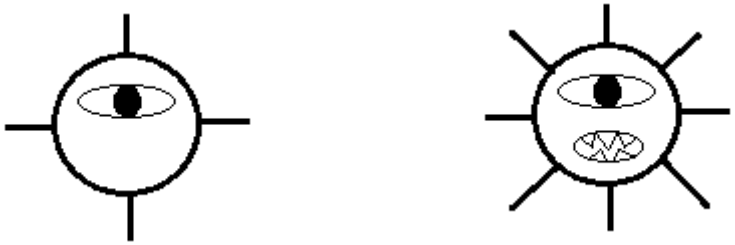
Example of a Cladogram



EXERCISE

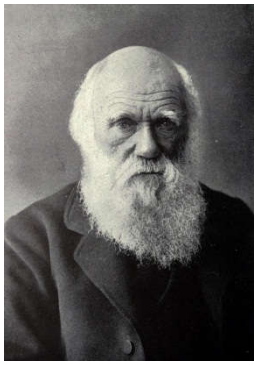


A B C D

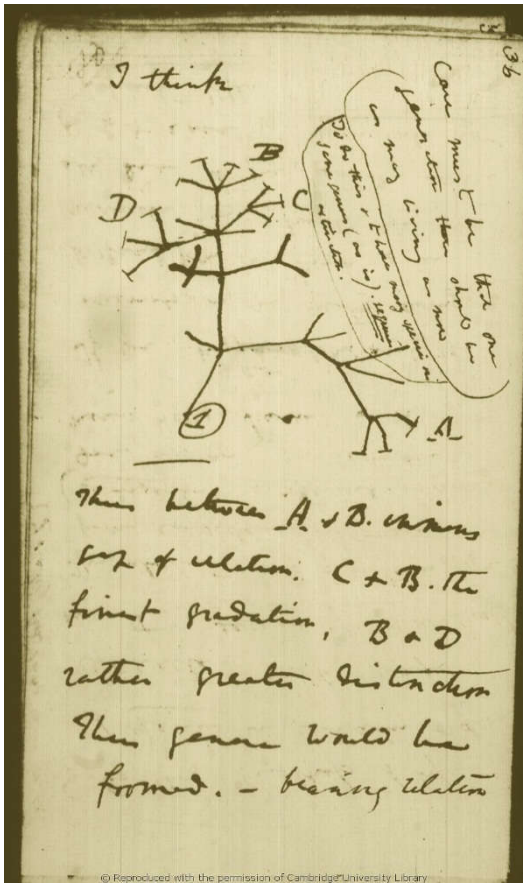


E F

Build an evolution cladogram of these 6 species?

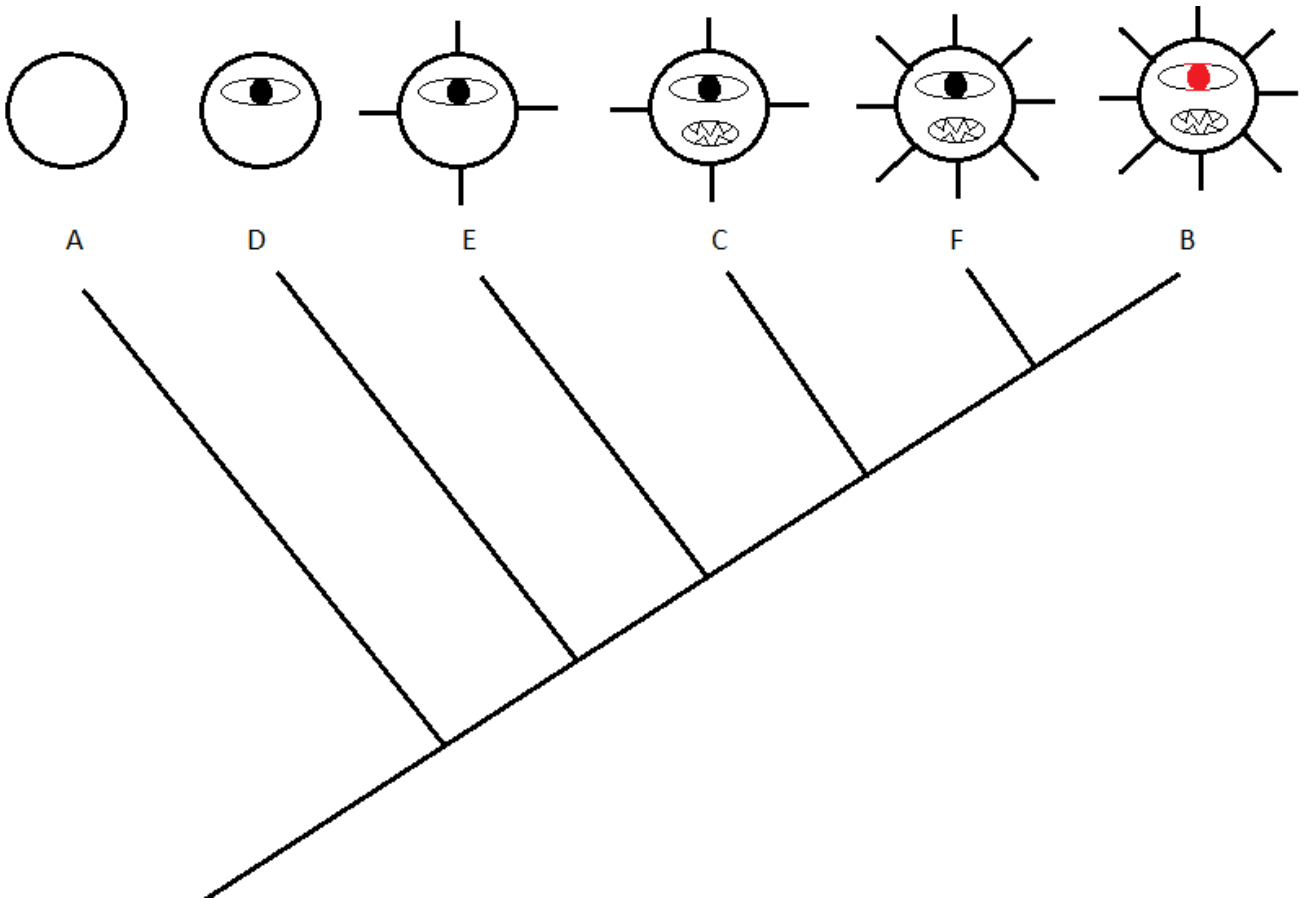


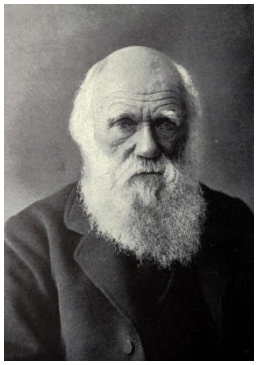
Charles Darwin
(1809-1882)



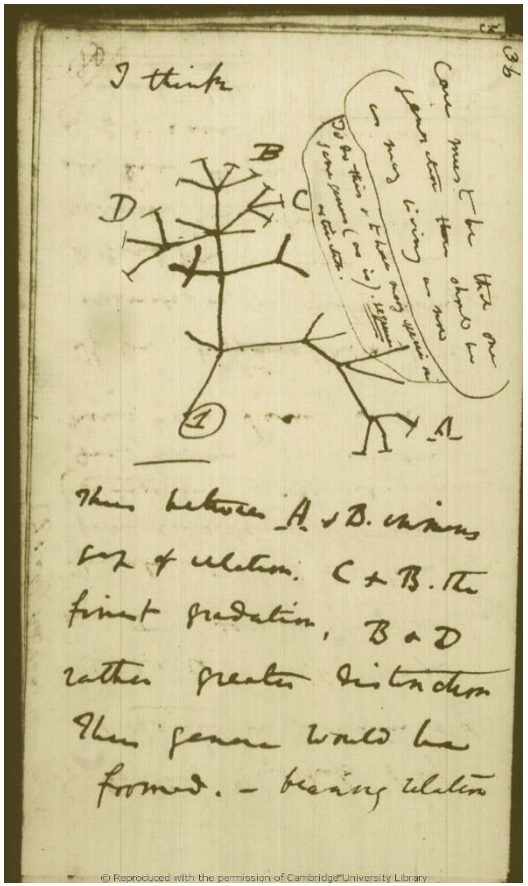
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ANSWER





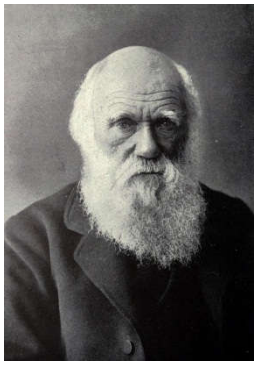
Charles Darwin
(1809-1882)



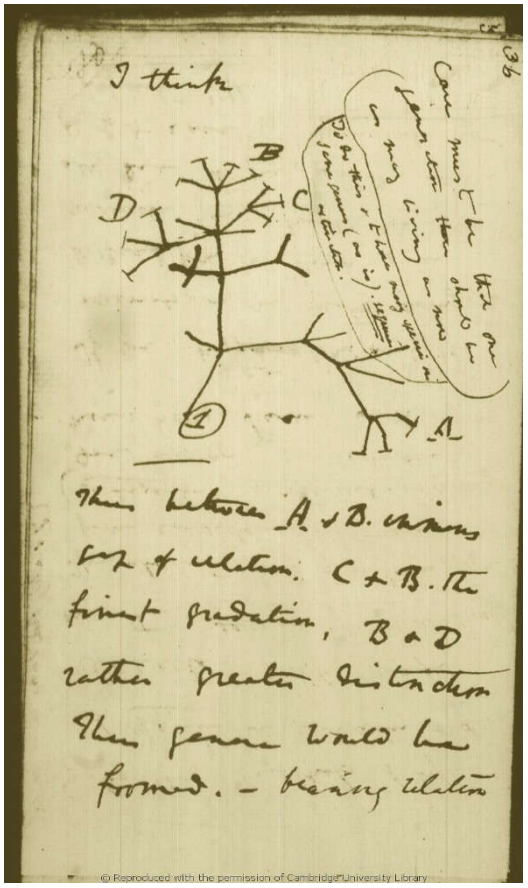
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Genetic mutations are raw material of evolutions

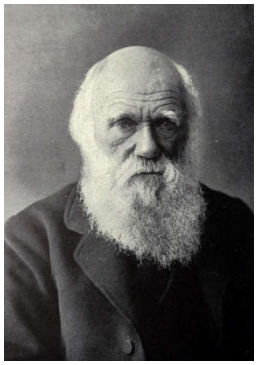




Charles Darwin
(1809-1882)



Genetic mutations are raw material of evolutions



Charles Darwin
(1809-1882)

Original DNA data:

Epinephelus miliarus	ATGAGATGACACGCTAACCCTGACCTTCCTGT
Epinephelus nigritus	ATGAAATGACACGCTAACTCTGACCTTCCTCT
Mycteroperca tigris	ATGAGATGACACGCTAACCCTGACCTTCCTCT
Plectropomus leopardus	ATGAAATGACACGCTAACTCTGACCCTGCTGTGCTCCCTGCCTCT
Plectropomus maculatus	ATGAGATGACACGCTAACCCTGACCTTGCTGTGCTCCCTTCTTTT
Variola louti	ATGAGATGACACGCTAACCCTGACCTTGCTGTGCTCCCTTCTTTT

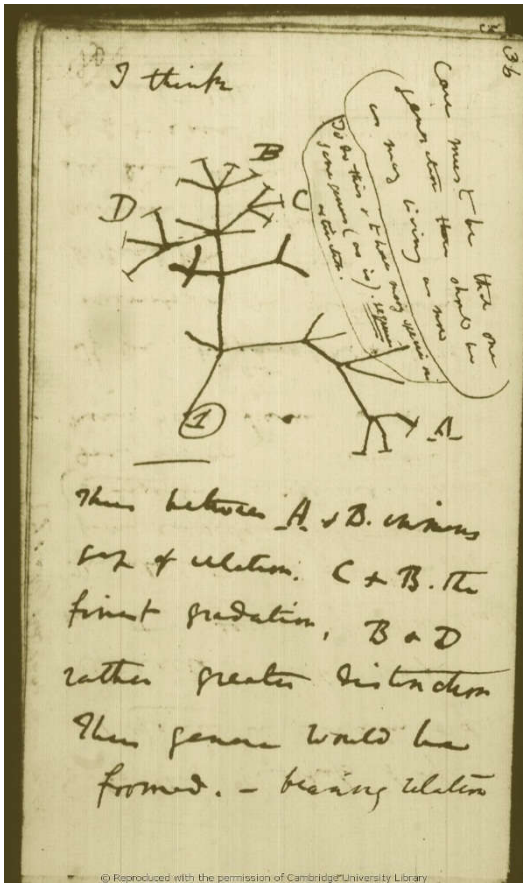
Data after alignment:

Epinephelus miliarus	ATGAGATGACACGCTAACCCTGACCTT----------CCTGT
Epinephelus nigritus	ATGAAATGACACGCTAACTCTGACCTT----------CCTCT
Mycteroperca tigris	ATGAGATGACACGCTAACCCTGACCTT----------CCTCT
Plectropomus leopardus	ATGAAATGACACGCTAACTCTGACCCTGCTGTGCTCCCTGCCTCT
Plectropomus maculatus	ATGAGATGACACGCTAACCCTGACCTTGCTGTGCTCCCTTCTTTT
Variola louti	ATGAGATGACACGCTAACCCTGACCTTGCTGTGCTCCCTTCTTTT

point substitutions deletion

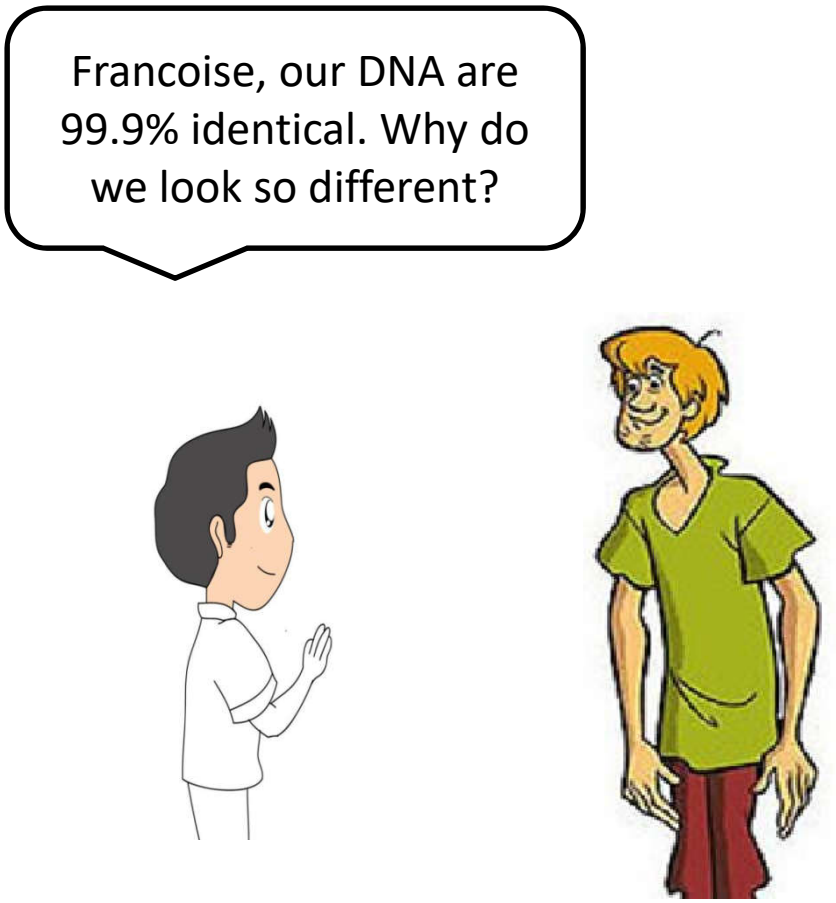
Mutations in DNA sequence from seven related species of tropical fishes (data are from intron 6 of *LDH-A* gene sequenced by the author from epinepheline serranids)

© 2011 Nature Education



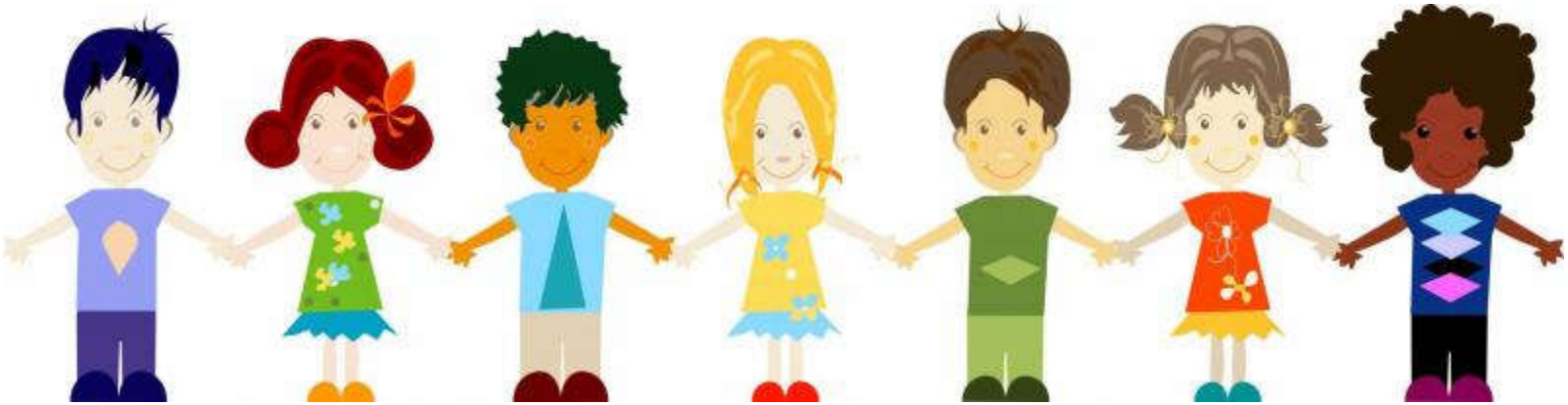
Genetic variation,
Genetic marker

Genetic variations underlie phenotypic differences



Genetic variations underlie phenotypic differences

All human beings differ from one another:
- Physical apperance: height, hair color, skin color…
- Susceptibility to disease
- Response to medication



The human genome is encoded as DNA within the 23 chromosome pairs , composed by **3 billions** base pairs. The human nucleotide diversity is estimated to be 0.1% to 0.4% of base pairs.

Genetic variations underlie phenotypic differences

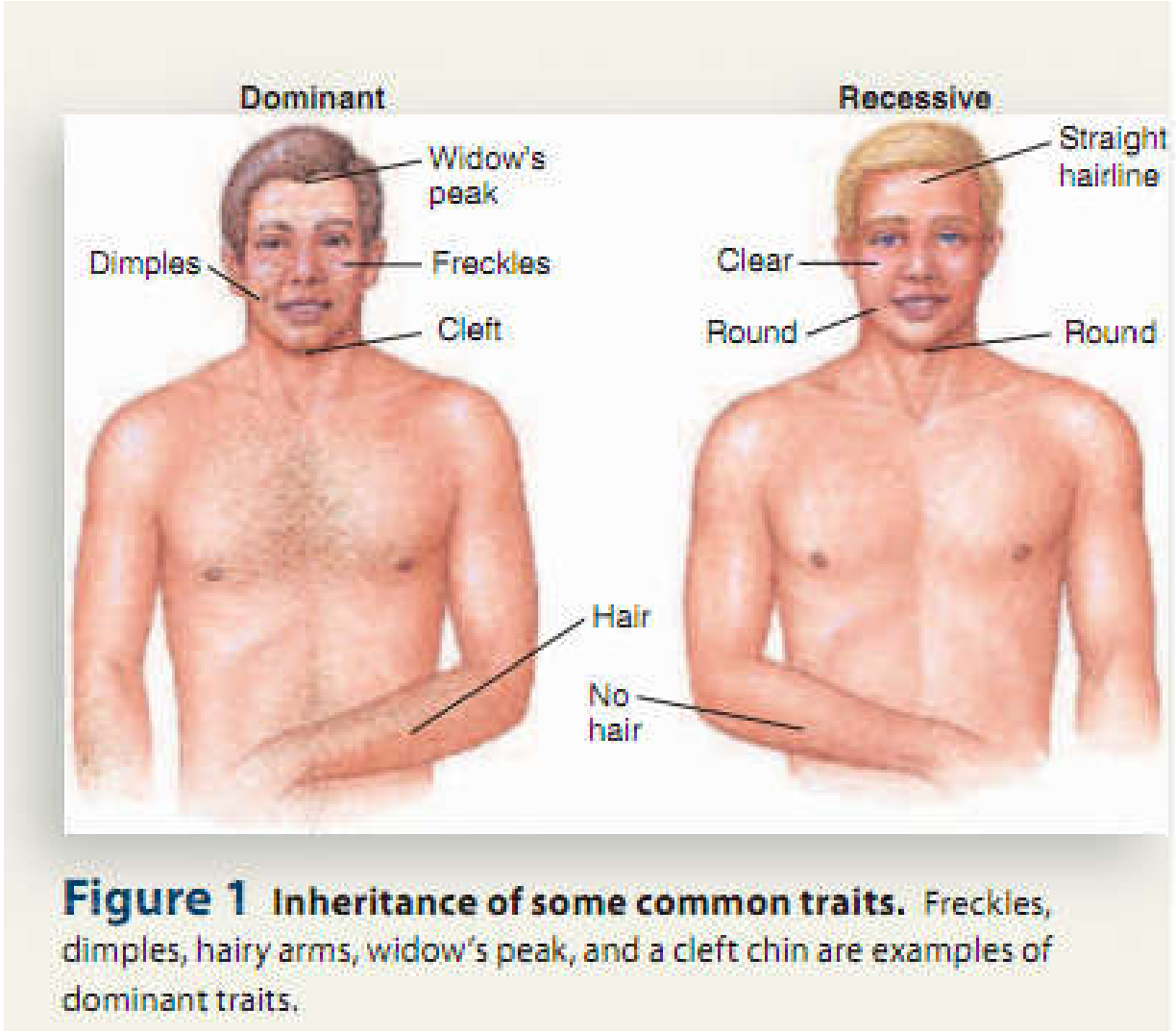
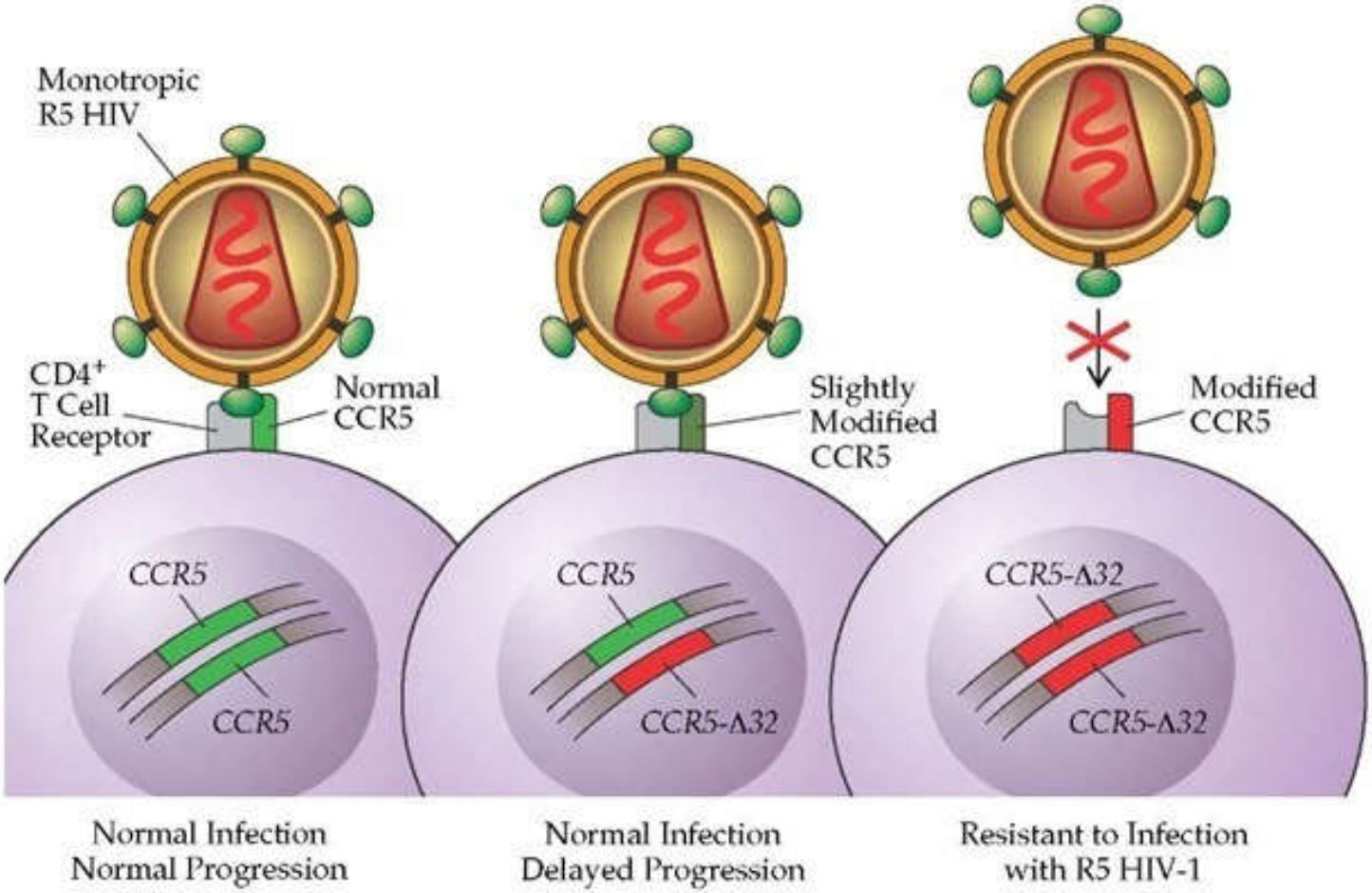


Figure 1 **Inheritance of some common traits.** Freckles, dimples, hairy arms, widow's peak, and a cleft chin are examples of dominant traits.

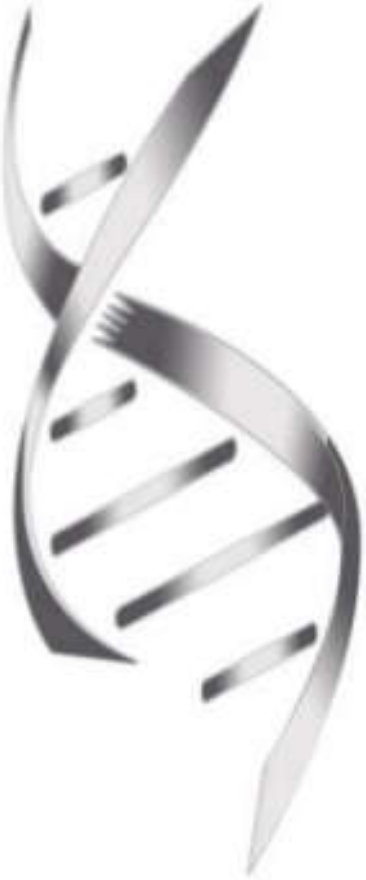
Genetic variations underlie phenotypic differences

Ex: Innate resistance to HIV



Genetic markers

- **RFLP** *Restriction fragment length polymorphism*
- **SSLP** *Simple sequence length polymorphism*
- **AFLP** *Amplified fragment length polymorphism*
- **RAPD** *Random amplification of polymorphic DNA*
- **VNTR** *Variable number tandem repeat*
- **SNP** *Single nucleotide polymorphism*
- **STR** *Short tandem repeat*
- **SFP** *Single feature polymorphism*
- **DArT** *Diversity Arrays Technology*
- **RAD** *markers Restriction site associated DNA markers*



SSR (Simple sequence repeat)

DNA markers which developed by amplifying microsatellite in the genome

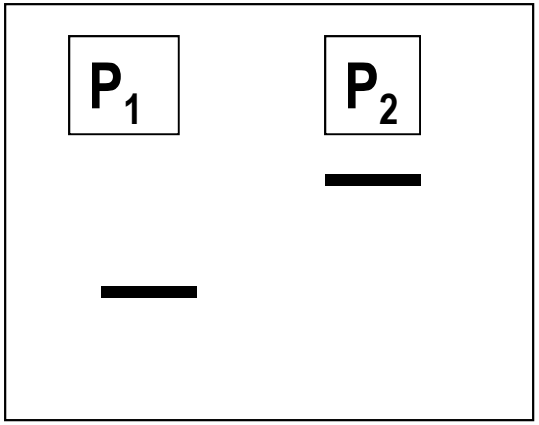
Sequence	Primer
ACTGTCG**ACACACACACACAC**GCTAGCT TGACAGC**TGTGTGTGTGTGT**G**C**GATCGA	$(AC)_7$
ACTGTCG**ACACACACACACACAC**GCTAGCT TGACAGC**TGTGTGTGTGTGTGT**G**C**GATCGA	$(AC)_8$
ACTGTCG**ACACACACACACACACACAC**GCTAGCT TGACAGC**TGTGTGTGTGTGTGTGTGT**G**C**GATCGA	$(AC)_{10}$
ACTGTCG**ACACACACACACACACACACACAC**GCTAGCT TGACAGC**TGTGTGTGTGTGTGTGTGTGTGT**G**C**GATCGA	$(AC)_{12}$

SSR polymorphisms

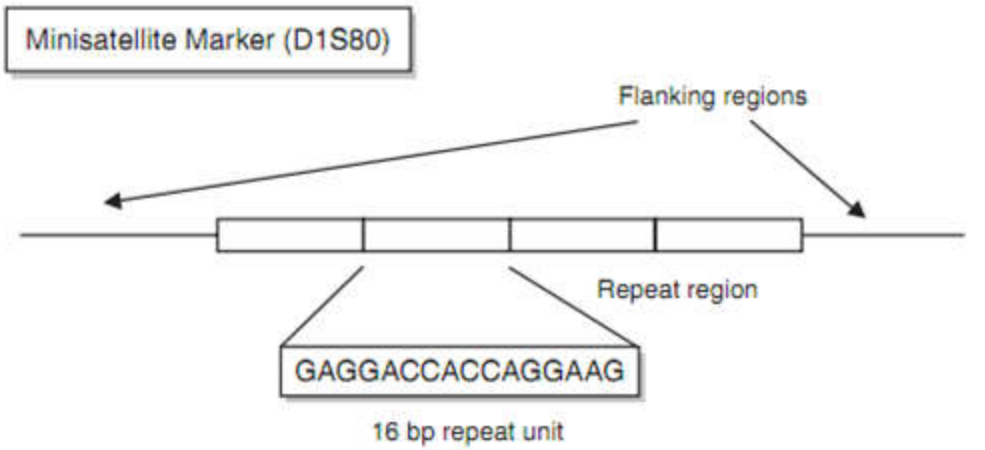
P_1 AATCCGGACTAGCTTCTTCTTCTTCTTCTTTAGCGAATTAGG

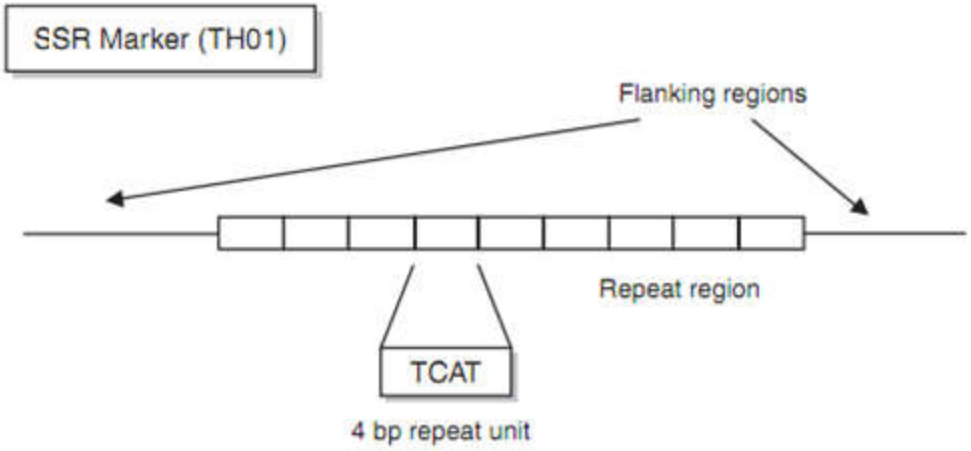
P_2 AAGGTTATTTCTTCTTCTTCTTCTTCTTCTTCTTAGGCTAGGCG

Gel configuration



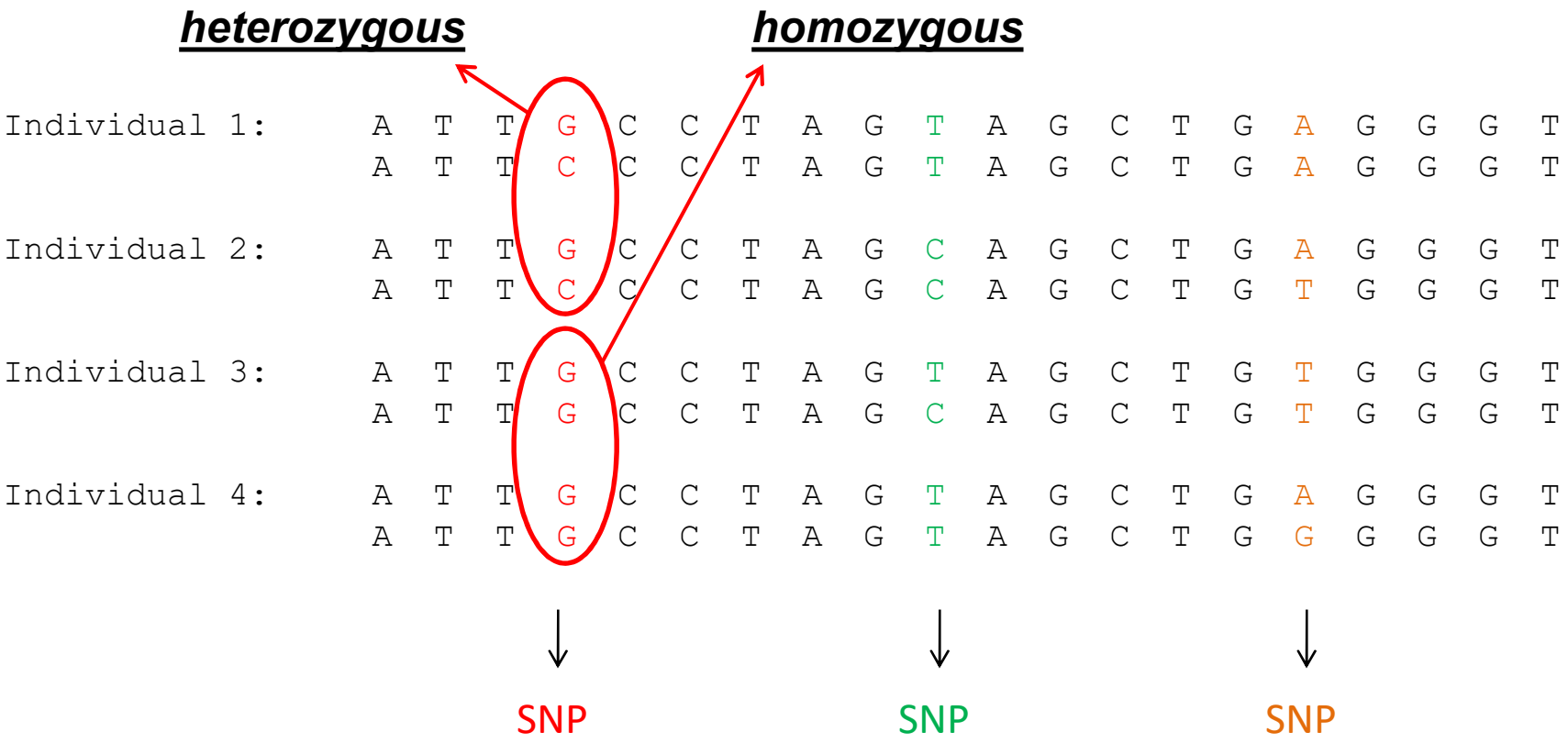
VNTRs


Minisatellite Marker (D1S80)
Flanking regions
Repeat region
GAGGACCACCAGGAAG
16 bp repeat unit


SSR Marker (TH01)
Flanking regions
Repeat region
TCAT
4 bp repeat unit

Single nucleotide polymorphism

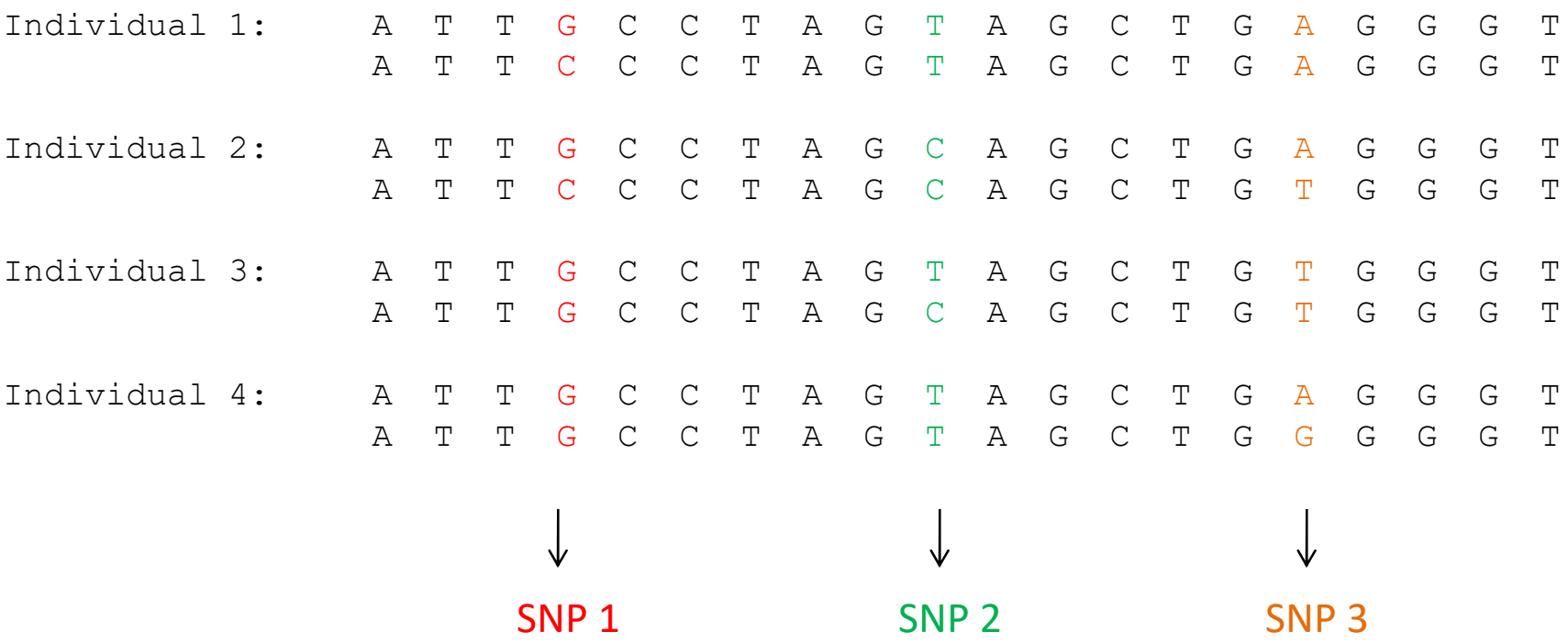
A **single nucleotide polymorphism** or , often abbreviated to just **SNP** is a variation in a single nucleotide that differs between members of a species or paired chromosomes in an individual.
SNPs are the most common type of sequence variation, estimated to account for 90% of all sequence variation.



Allele frequency: for SNP 1

%G = 6G/(6G+2C) = 0.75 ⟶ Major allele

% C = 2C/(6G+2C) = 0.25 ⟶ Minor allele



Allele frequency: for SNP 1

%G = 6G/(6G+2A) = 0.75 ⟶ Major allele

% C = 2A/(6G+2A) = 0.25 ⟶ Minor allele

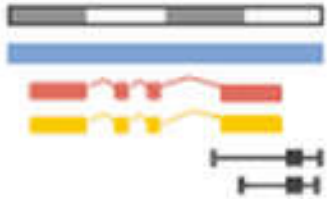
- ***Minor allele frequency (MAF):*** the MAF is actually the second most frequent allele value.

- MAF distinguish common polymorphism from rare variants (MAF<0.01)

MAF for SNP 1 is 0.25

Ensembl Features

- Gene builds for ~70 species
- Gene trees
- Regulation data
- Variation display and tools
- BioMart (data export)
- Display of user data
- Display of External Data (DAS)
- Programmatic access via the Perl API
- Completely Open Source











Ensembl (http://asia.ensembl.org/index.html)

